

1. Introduction to Augmented Reality (0–10 minutes)

Definition of Augmented Reality

Augmented Reality (AR) is a technology that **enhances the real-world environment by overlaying digital information** such as text, images, audio, or 3D objects in real time.

Key Characteristics of AR:

- Combines real and virtual content
- Real-time interaction
- Accurate 3D registration
- Works in real-world environments



Examples:

Snapchat filters, Google Maps Live View, IKEA Place

2. Types of Augmented Reality (10–30 minutes)

A. Marker-Based AR (10 minutes)

Description:

Uses predefined visual markers (QR codes, black-and-white patterns) to trigger digital content.

Working Principle:

Camera detects marker → identifies pattern → overlays virtual object.

Examples:

AR textbooks, product packaging AR

Advantages:

- High accuracy
- Easy implementation

Limitations:

- Requires visible marker
- Limited flexibility

B. Marker-less AR (10 minutes)

Description:

Does not require markers; uses environmental features.

Technologies Used:

- SLAM (Simultaneous Localization and Mapping)
- Depth sensors
- Computer vision

Examples:

AR furniture placement apps, indoor navigation

Advantages:

- More natural interaction
- Works in dynamic environments

Limitations:

- Computationally intensive
 - Requires powerful hardware
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C. Location-Based AR (10 minutes)

Description:

Uses geographical data to overlay virtual content based on user location.

Technologies Used:

- GPS
- Compass
- Accelerometer

Examples:

Pokémon Go, location-based tourism apps

Advantages:

- Outdoor usability
- Scalable

Limitations:

- GPS accuracy issues
 - Limited indoor usage
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3. AR Pipeline (30–45 minutes)

Definition:

The AR pipeline is a **sequence of processes** that enables digital content to be accurately placed over the real world.

Steps in AR Pipeline:

1. **Scene Capture** – Camera captures real-world view
2. **Tracking & Registration** – Determines position and orientation
3. **Processing & Rendering** – Generates virtual content
4. **Display** – Combines real and virtual images

 **Explain with block diagram**

4. Tracking Methods in AR (45–55 minutes)

Common Tracking Techniques:

- **Marker-based Tracking**
- **Feature-based Tracking**
- **Sensor-based Tracking**
- **Hybrid Tracking**

 **Comparison of tracking accuracy and use cases**

5. Recap & Interaction (55–60 minutes)

Key Takeaways:

- AR enhances reality, not replaces it
- Different AR types serve different applications

- Tracking accuracy is critical in AR
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Important Questions (Exam-Oriented)

1. **Explain the different types of Augmented Reality with suitable examples.**
(8–10 marks)
2. **Describe the Augmented Reality pipeline with a neat diagram.**
(8 marks)
3. **Discuss various tracking methods used in Augmented Reality systems.**
(5 marks)